

assignment5 voltage, electric potential

 This is a preview of the published version of the quiz

Started: Mar 22 at 6:49pm

Quiz Instructions

You have excellent tutorials for almost all of them. Don't look at the solutions unless you try really hard. Pause the videos.

The last attempt counts. 5 min to complete an assignment is suspicious. 50min is more like it

Make sure to open study the textbook first.

You can start and save the work.



Question 1 10 pts

Points A [at (2, 3) m] and B [at (5, 7) m] are in a region where the electric field is uniform and given by $E = 4\mathbf{i} + 3\mathbf{j}$ N/C. or $E(4,3)$ N/C. What is the potential difference $V_A - V_B$? Try on your own first.

hint: Remember the definition of $V_A - V_B$. If E is constant then

$[-E \cdot \Delta S = V_A - V_B]$ (slide 42) Its a dot product between E and the vector displacement.

dot product is $(x_1 x_2 + y_1 y_2 + z_1 z_2)$. Trace the vectors E and AB you see E are both pointing up. So $V_A > V_B$.

E shows the lower potential. Now there is an **EXCELLENT** tutorial in the dropbox.



33V



11V



24V



27V



Question 2 10 pts

A proton (mass = $1.67 \cdot 10^{-27}$ kg, charge = $1.60 \cdot 10^{-19}$ C) moves from point A to point B under the influence of an electrostatic force only. At point A the proton moves with a speed of 50 km/s. At point B the speed of the proton is 80 km/s. Determine the potential difference $V_B - V_A$.

hint: there is a video. yey

☐

20V

☐

-27V

☐

-20V

☐

-40V

☒

Question 3 10 pts

A proton ($m = 1.7 \cdot 10^{-27}$ kg, $q = +1.6 \cdot 10^{-19}$ C) starts from rest at point A and has a speed of 40 km/s at point B. Only electric forces act on it during this motion. Determine the electric potential difference $V_B - V_A$.

hint: excellent tutorial in dropbox

☐

-17V

☐

-4.8V

☐

-8.5V

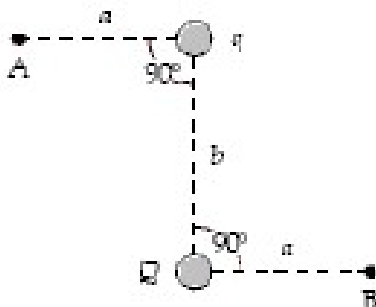
☐

8.5V

☒

Question 4 10 pts

Point charges q and Q are positioned as shown. If $q = +2.0$ nC, $Q = -2.0$ nC, $a = 3.0$ m, and $b = 4.0$ m, what is the electric potential difference, $V_A - V_B$?



hint: video

☐

0V

☐

4.8V

☐

7.2V

☐

8.4V



Question 5 10 pts

A $+4.0\text{-mC}$ charge is placed on the x axis at $x = +3.0$ m, and a -2.0-mC charge is located on the y axis at $y = -1.0$ m. Point A is on the y axis at $y = +4.0$ m. Determine the electric potential at point A (relative to zero at the origin).

V_A .

hint: video

☐

3.6kV

☐

4.8kV

☐

9.6kV

☐

6 kV



Question 6 10 pts

As a proton moves in a direction perpendicular to the electric field lines

☐

both its electric potential and electric potential energy remain constant.

☐

it is moving from high potential to low potential and gaining electric potential energy.

☐

it is moving from high potential to low potential and losing electric potential energy.

☐

it is moving from low potential to high potential and losing electric potential energy.

☐

it is moving from low potential to high potential and gaining electric potential energy.



Question 7 10 pts

A charge of 10 nC is distributed uniformly along the x axis from $x = -2$ m to $x = +3$ m. Which of the following integrals is correct for the electric potential (relative to zero at infinity) at the point $x = +5$ m on the x axis?

he y axis at $y = +4.0$ m. Determine the electric potential at point A (relative to zero at the origin).

fl V_A .

hint: video. excellent tutorial. Teach you how to use calculus skills (therefore the calc in W calc Physics)

A, B, C, D

☐ $\int_{-2}^3 \frac{18dx}{5-x}$

☐ $\int_{-2}^3 \frac{18dx}{x}$

☐ $\int_{-2}^3 \frac{50dx}{5-x}$

☐ $\int_{-2}^3 \frac{50dx}{5+x}$

☐

C

☐

A

☐

B

☐

D

⋮

Question 8 10 pts

A rod (length = 2.0 m) is uniformly charged and has a total charge of 5.0 nC. What is the electric potential (relative to zero at infinity) at a point which lies along the axis of the rod and is 3.0 m from the center of the rod?

hint: video

☐

22V



25V



12V



16V



Question 9 10 pts

A solid conducting sphere (radius = 5.0 cm) has a charge of 0.25 nC distributed uniformly on its surface. If point A is located at the center of the sphere and point B is 15 cm from the center, what is the magnitude of the electric potential difference between these two points?

hint: video



45V



30V



25V



60V



Question 10 10 pts

As an electron moves in the direction the electric field lines



both its electric potential and electric potential energy remain constant.



it is moving from low potential to high potential and gaining electric potential energy.



it is moving from high potential to low potential and gaining electric potential energy.



it is moving from high potential to low potential and losing electric potential energy.



it is moving from low potential to high potential and losing electric potential energy.

Quiz saved at 6:49pm

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